

Technology Stack

Date	27 June 2026
Team ID	
Project Name	Rising Waters-ML Flood Prediction System
Maximum Marks	2 Marks

Technology Stack

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	Provides a simple and interactive web interface for users to enter weather parameters and view flood prediction results.
2	Backend / Server-Side	Python, Flask Framework	Handles user requests, performs data preprocessing, loads the trained XGBoost model, and returns prediction results through the web application.
3	Database / Data Storage	Kaggle Flood Prediction Dataset, Joblib/Pickle (<code>floods.save</code>)	Stores the dataset used for model training and the trained machine learning model for prediction. <i>(If you use SQLite/MySQL, mention it here instead.)</i>
4	Cloud / Hosting / Deployment	IBM Cloud	Hosts and deploys the Flask web application, making the flood prediction system accessible online.
5	Version Control & CI/CD	Git and GitHub	Used for source code management, version tracking, and collaboration during development.
6	Third-Party APIs / Other Tools	Anaconda Navigator, Jupyter Notebook, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, XGBoost	Used for development, data preprocessing, visualization, machine learning model training, evaluation, and analysis.